

Luke Serrano

✉ lukeserrano0@gmail.com | [in](#) Luke Serrano | 📞 (618) 334-5046

EXPERIENCE

Sandia National Laboratories Year-Round Intern (L Clearance) Jan 2024 - Present

Navigation, Guidance, and Control (NGC) Engineer Albuquerque, NM

- Analyzed flight navigation software and created an independent inertial navigation algorithm in an attempt to resolve a non-conformance from system testing
- Improved navigation verification framework to ensure robustness and reliability of current and future missile navigation algorithms by adding the ability to play-back telemetry data in simulation
- Created a valve pressure analysis script to verify valve functionality and characterize valve delay timings for simulation, has been implemented into attitude control system (ACS) test procedures
- Modified auto-generated XML messaging framework, C++ embedded, and Java interface software to set ACS algorithm parameter in real-time for dynamic initial tank pressure testing
- Proved ability to work within multi-language software framework, dive into mathematical theory, develop within a Git workflow, and contribute meaningfully to an NGC team

Missouri S&T Satellite Research Team (M-SAT) Aug 2022 - Present

Missouri University of Science & Technology Rolla, MO

- Integrating NASA's C programming based 42 simulator and spacecraft NGC algorithms to verify requirements via simulation
- Manage NGC team Git repositories and simulation development tasking
- Troubleshoot SPI, I2C, and USART communication on PCB's to develop interface between microcontrollers and sensors
- Conducted system-level testing to verify final hardware and software before flight model integration of team's first launched satellite

Northrop Grumman Internship June 2023 - Aug 2023

Northrop Grumman Space Sector Gilbert, AZ

- Systems Safety and Reliability Engineering Intern at satellite manufacturing facility
- Exposure to industry with requirements, a paying customer, deadlines, and expected mission success

Orbit Determination and Estimation Class Project Aug 2024 - Present

- Developed spacecraft error-state multiplicative extended Kalman filter utilizing IMU, GPS, and magnetometer sensors
- Actively verifying convergence in NASA's 42 simulator

EDUCATION

Missouri University of Science & Technology Fall 2022 - Spring 2025

B.S. in Aerospace Engineering **GPA: 3.8**

Orbit Determination and Estimation; Optimal Control and Estimation; Orbital Mechanics II;

SKILLS

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|---------------------------------|-----------------------|--------------|
| - C++ | - MATLAB | - Java |
| - Git Development | - Quick Learner | - Leadership |
| - Circuitry Board Communication | - Strong Communicator | - Discipline |